Abstract

In this talk I would like to present you the main results of my PhD thesis. They encompass the formal definition of mutual simulation of Hamiltonians and the development of a theory investigating the computational resources required for simulating Hamiltonians.

Mutual simulation of Hamiltonians builds upon a control theoretic model of quantum computers. In this model unitary transformations are implemented by performing an external control operation, letting the system evolve for a certain time, and repeating both steps several times. We study the ability of the system to implement transformations representing the time evolution of another quantum system. Considering the cost of simulating the time evolution of a quantum system by a given one defines in a certain sense the comparison of the "computational power" of different systems. The determination of the cost in general is a computationally hard problem. But it can be solved to a large extent if we consider infinitesimal time intervals only. This leads to our definition of mutual simulation of Hamiltonians.

We derive lower bounds on the simulation complexity. An important tool for obtaining these bounds is the characterization of Hamiltonians by graphs; the graphs represent the coupling topology of the Hamiltonians, that is, which pairs of subsystems interact. Our bounds show that the simulation complexity depends strongly on the properties of these graphs. We develop optimal simulation algorithms based on methods of graph theory, representation theory of finite groups, design theory, and convex optimization theory. Furthermore, we show how the elementary quantum gates of the quantum circuit model can be implemented within the control theoretic model. This shows an interesting connection between the quantum circuit model and the control theoretic model.

We show as an interesting application of our simulation methods that they allow an efficient realization of adiabatic quantum computing in the control theoretic model. It is important that only physically realistic Hamiltonians (that is, with a local coupling topology) are used as computational resources. We construct such Hamiltonians whose energy minima encode the solutions of the NP-complete problem "max independent set". Due to their special structure these Hamiltonians can be simulated efficiently by Ising-interactions on a two-dimensional lattice.